

Stage 12: Order Management System (OMS) & Multi-Vendor Sub-Orders

1. Overview & Architecture

Stage 12 upgrades the Alight International Marketplace with a production-grade Order Management System (OMS) featuring automatic multi-vendor sub-order splitting, vendor-level fulfillment workflows, split shipment tracking, automated commission/payout calculations, and master order state synchronization driven by Spring domain events.

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flowchart TD
    Cart[User Cart / Guest Session] -->|Checkout Initiated| CS[CheckoutService]
    CS --> StockRes[Confirm Stock Reservation]
    CS --> MasterOrd[Create Master Order\n'ORD-YYYY-XXXXX']
    CS --> Split[Group Items by Vendor]
    Split --> SubOrd1[VendorOrder 1\n'ORD-...-V1'\nCommission & Payout]
    Split --> SubOrd2[VendorOrder 2\n'ORD-...-V2'\nCommission & Payout]
    MasterOrd --> Evt1[OrderCreatedEvent]
    SubOrd1 --> Fulfill1[Vendor 1 Updates Status\nPROCESSING -> SHIPPED -> DELIVERED]
    SubOrd2 --> Fulfill2[Vendor 2 Updates Status\nPROCESSING -> SHIPPED -> DELIVERED]
    Fulfill1 & Fulfill2 --> Sync[Master Order Status Sync Engine]
    Sync --> Evt2[OrderStatusUpdatedEvent]
```

2. Key Components & Implementation Details

A. Flyway Schema Migration (`V38__order_management_and_suborders.sql`)

- Added commission and payout auditing columns:
 - `commission_rate NUMERIC(5, 2)` : Vendor commission rate snapshot. - `commission_amount NUMERIC(12, 2)` : Calculated platform commission. - `payout_amount NUMERIC(12, 2)` : Net payout to vendor (`grandTotal - commission`). - `notes TEXT` : Vendor/fulfillment notes.
- Performance Indexes:
 - `idx_orders_user_created` on `orders(user_id, created_at DESC)` -
 - `idx_vendor_orders_vendor_status` on `vendor_orders(vendor_id, fulfillment_status)` -
 - `idx_vendor_orders_master_order` on `vendor_orders(master_order_id)` -
 - `idx_order_items_vendor_order` on `order_items(vendor_order_id)` -
 - `idx_order_items_product_variant` on `order_items(product_id, variant_id)`

B. Multi-Vendor Sub-Order Splitting & Commission Engine

- Checkout Processing (`CheckoutServiceImpl.java`):

- Automatically groups cart items by vendor. - Calculates line item subtotal, GST / tax rates (CGST/SGST vs IGST based on vendor state and shipping destination), and shipping allocation. - Computes vendor commission and net vendor payout amount. - Generates unique sub-order identifiers (`ORD-YYYY-XXXX-V1` , `ORD-YYYY-XXXX-V2`). - Publishes `OrderCreatedEvent` .

C. Sub-Order Fulfillment Lifecycle & Master Order State Synchronization

- **Fulfillment Engine (`OrderServiceImpl.java`):**
 - `updateVendorOrderFulfillment` : Vendors update their sub-order status with courier name, tracking number (AWB), and notes. Automatically captures `shippedAt` and `deliveredAt` timestamps. - **Auto-Progression Logic:** - If any vendor order moves to `PROCESSING` , master order moves to `PROCESSING` . - If all vendor orders are `SHIPPED` or `DELIVERED` , master order automatically transitions to `SHIPPED` . - When all vendor orders reach `DELIVERED` , master order transitions to `DELIVERED` . - Dispatches `OrderStatusUpdatedEvent` for audit trails, analytics, and customer notifications.

D. Security & Data Isolation

- RBAC validation ensures buyers can only access their own orders.
- Vendors can only view and fulfill sub-orders belonging to their approved store.
- Administrators retain cross-tenant visibility and manual status override capabilities.

3. Verification & Test Suite

- Comprehensive end-to-end integration test suite implemented in `OrderManagementIntegrationTest.java` :
 1. `testMultiVendorOrderCheckoutAndSubOrderSplitting` : Multi-vendor checkout, sub-order splitting, commission and payout calculation, address linking.
 2. `testVendorOrderFulfillmentProgression` : Full sub-order fulfillment lifecycle (`PROCESSING` → `SHIPPED` with AWB → `DELIVERED`) and automatic master order status synchronization.
 3. `testOrderAccessControlAndFiltering` : Multi-tenant isolation between buyers, vendors, and administrators.
- Unit test suites updated: `CheckoutServiceTest.java` and `OrderServiceTest.java` .
- Build compilation: `BUILD SUCCESS` across 56 test classes.